



Choice-Head Distillation for Efficient Dental MCQ Answering

Decision-Space Distillation for Deployable Medical QA

DeepSeek-V3 Teacher | Qwen2.5-14B Student | CMExam Full-Test 991 Questions

Key Result: 89.10% student accuracy > 87.18% teacher accuracy

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Presentation Outline

1. Motivation and problem setting
2. Research questions and Choice-Head method
3. Why decision-space distillation matters
4. Experimental setup and benchmark scale
5. Main results and student-over-teacher finding
6. Contributions, discussion, and conclusion

Goal: preserve exam performance while making dental MCQ systems smaller, cheaper, and easier to deploy.

Current Tension

Medical LLMs are accurate on exam benchmarks
But inference cost and hardware demand remain high
Deployment value matters beyond raw benchmark scores

Task Structure

Dental QA is a five-option MCQ task
Decision space is small and well-structured
Open-ended generation is not required here

Mismatch

Full-vocabulary distillation is redundant
API teachers rarely expose internal logits
Task-aligned supervision is needed

Research Questions

RQ1

What to Distill?

Whole vocabulary or only
A/B/C/D/E?

RQ2

Can Students Exceed Teachers?

Under a task-aligned objective

RQ3

Can It Stay Black-Box Friendly?

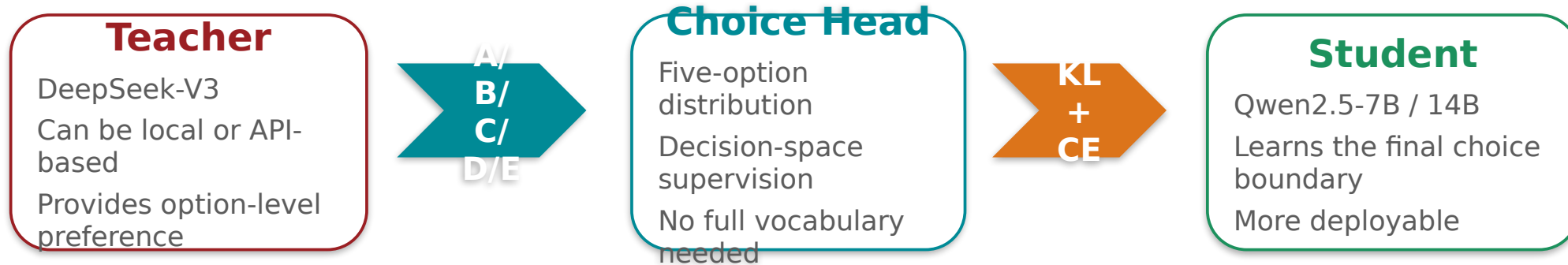
Compatible with API teachers

Central research question

For a five-option medical exam task, the important transfer target may be the decision structure rather than the full language-model distribution.

If the supervision target is redesigned around the downstream task, smaller students may become both deployable and unexpectedly strong.

03 Core Method: Choice-Head Distillation



Stage 1 objective

$$L = \alpha * KL(p_T || p_S) + (1 - \alpha) * CE$$

The student learns relative preferences across the five answer options instead of the entire token vocabulary.
Main setting: one Stage 1 epoch with $\alpha = 0.35$ for the best 14B run.

Design intuition

Keep the part of teacher uncertainty that directly affects the final option.
Drop supervision dimensions that do not matter for a five-choice decision.

This is why the method stays lightweight while remaining competitive.

Why Choice-Head Matters

Alignment

Task-Specific

Supervision follows the five-option structure

Efficiency

Lower Cost

Less redundant training and memory usage

Teacher Access

Black-Box Friendly

Works with API-only teachers

Transfer Signal

Useful Uncertainty

Preserves option-level confusion structure

The method changes distillation from a generic language-model compression problem into a task-specific decision transfer problem.

Better aligned supervision often matters more than a more complicated pipeline.
For structured MCQs, the decision-space target is a cleaner and more deployable abstraction.

Experimental Setup

Dataset

**6,591
Questions**

CMExam-based single-
choice resplit

Splits

**4608 /
991 / 991**

Train / Val / Test

Students

**Qwen2.5
7B / 14B**

LoRA fine-tuning

Teacher

**DeepSeek-
V3**

87.18% on the 991-
question test

Evaluation scope

Main benchmark: 991-question full-data
test set

Dental subset retained for specialty-
focused validation

Large-scale resplit makes the main
conclusion more stable

Training setup

LoRA rank = 16, LoRA alpha = 32

Best 14B setting uses one Stage 1
epoch

Main learning rate: 1e-4

Stage 2 is tested but not required for the
best 14B result

Main Results

Teacher**87.18%**

DeepSeek-V3 on full-data test

14B Zero-Shot**83.55%**

Base student without distillation

14B Distilled Mean**88.67%**

Three-seed average

14B Distilled Best**89.10%**

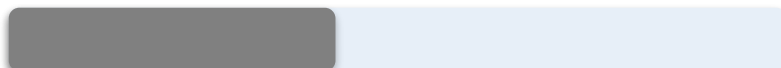
Best single run

Teacher



87

14B Zero-shot



83

14B Mean



88

14B Best



89

Takeaway

The distilled student gains 5.12 percentage points over the 14B zero-shot baseline on average. The best 14B student exceeds the teacher on the same 991-question benchmark.

The 7B student also improves str
tie

+1.92 points vs teacher

Two Main Contributions

Contribution 1: Method

Introduces a task-aligned decision-space distillation framework for five-option medical MCQs.

Replaces generic vocabulary-space supervision with a cleaner option-level target.

Keeps the framework practical for both local and API teachers.

Contribution 2: Empirical Finding

Shows that a smaller student can outperform a stronger teacher under the task-aligned formulation.

Validates the result on a 991-question test set rather than only a tiny benchmark slice.

Frames the student-over-teacher effect as a consequence of better task alignment, not a distillation failure.

Stage 2 is Not Always Better

Helpful for some smaller students
Can mildly hurt a strong 14B student
Extra GT fine-tuning may erase soft-label structure

Why the Student Can Win

Teacher soft labels provide structure
Student is still optimized for the benchmark
Task-aligned supervision sharpens the decision boundary

Broader Message

More complexity is not the point
The supervision target must fit the task
Decision-space transfer can be a practical design rule

For structured medical multiple-choice tasks, the critical design choice is not how much of the teacher to copy, but which part of the teacher signal actually matters.

Choice-Head distillation makes smaller dental MCQ models both practical and highly competitive.

Choice-Head supervision is an effective alternative to full-vocabulary distillation for structured MCQs. The student reaches 89.10% and surpasses the 87.18% DeepSeek-V3 teacher on the 991-question test set. The model stays lightweight, deployment-oriented, and compatible with black-box teachers.

Thank you. Questions are welcome.

Why distill only five options?

The downstream task is a five-option decision problem.

Full-vocabulary supervision introduces unnecessary dimensions.

Option-level transfer is more task-aligned and more efficient.

Why can the student outperform the teacher?

The student learns teacher soft-label structure.

It is still optimized directly for the benchmark objective.

Task-aligned supervision can yield a stronger decision boundary.

Can this extend to open-ended QA?

Not directly.

The current method is built for structured MCQs.

Open-ended QA would require a different supervision target.